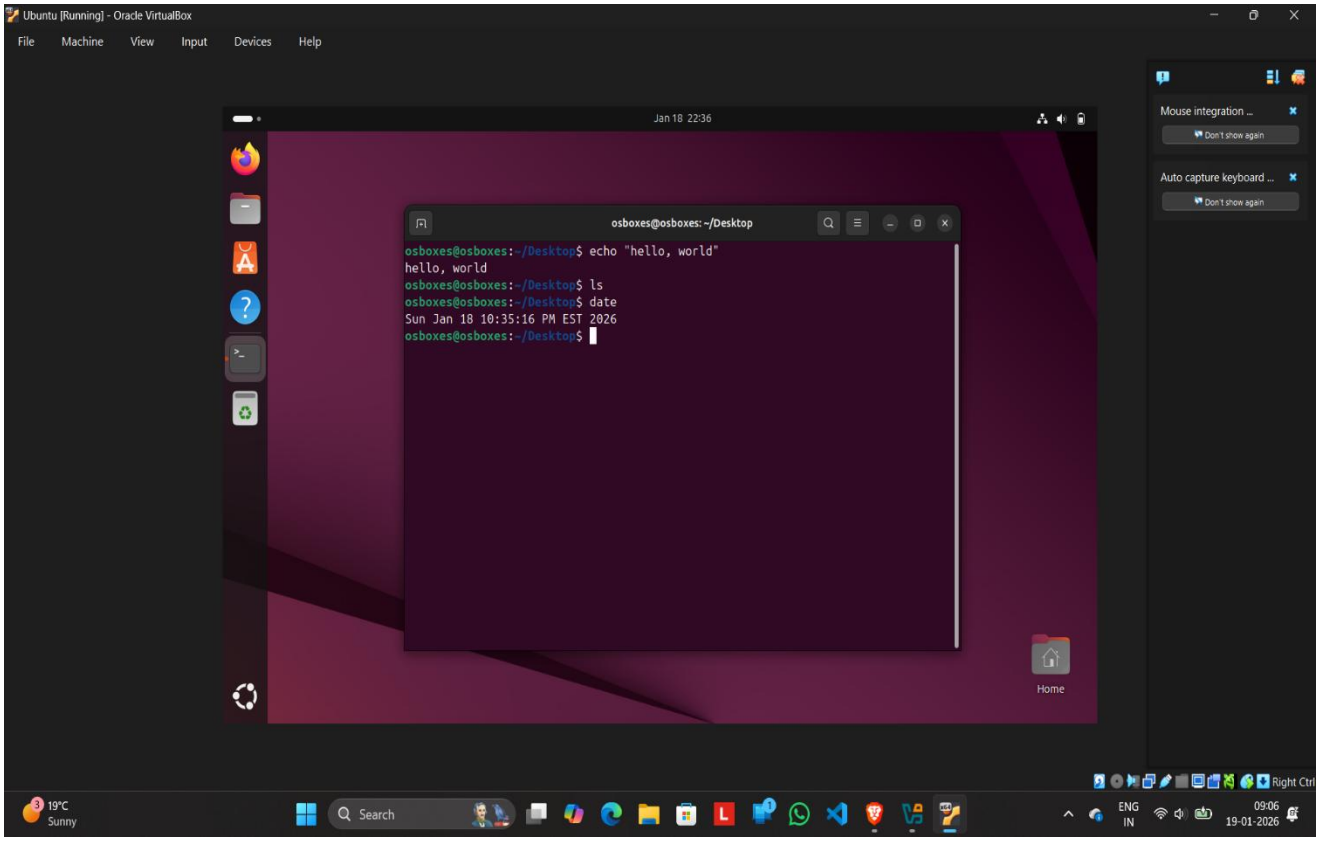
Cloud Computing

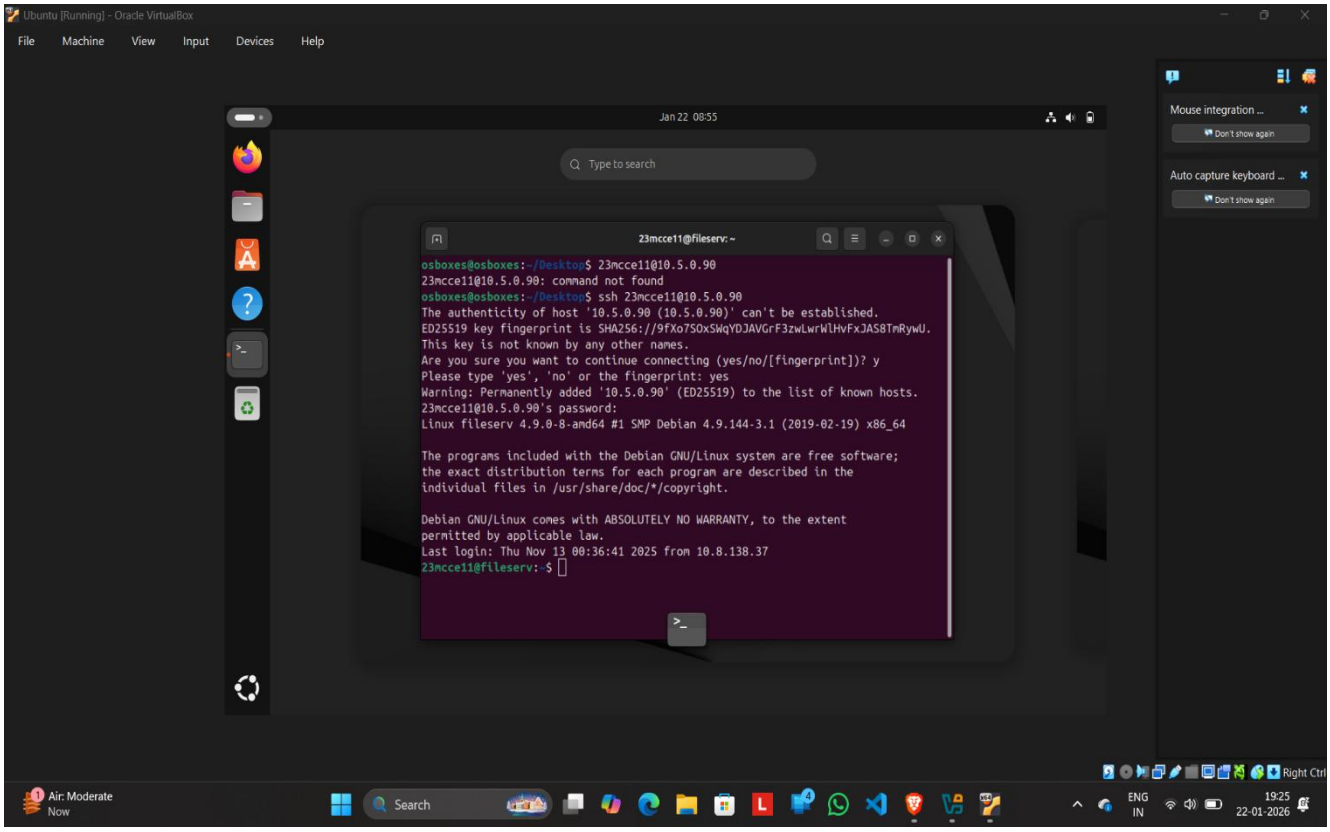
Assignment-1

Q1 :

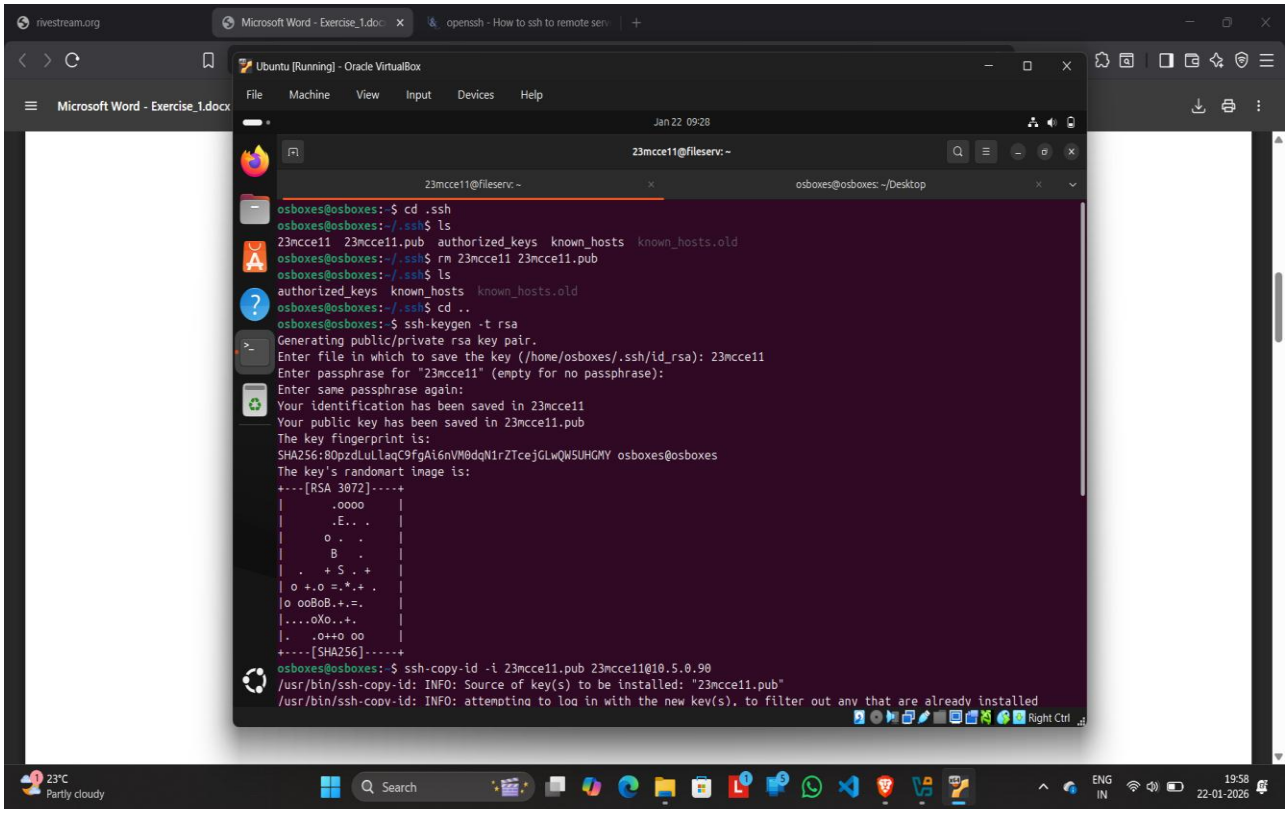


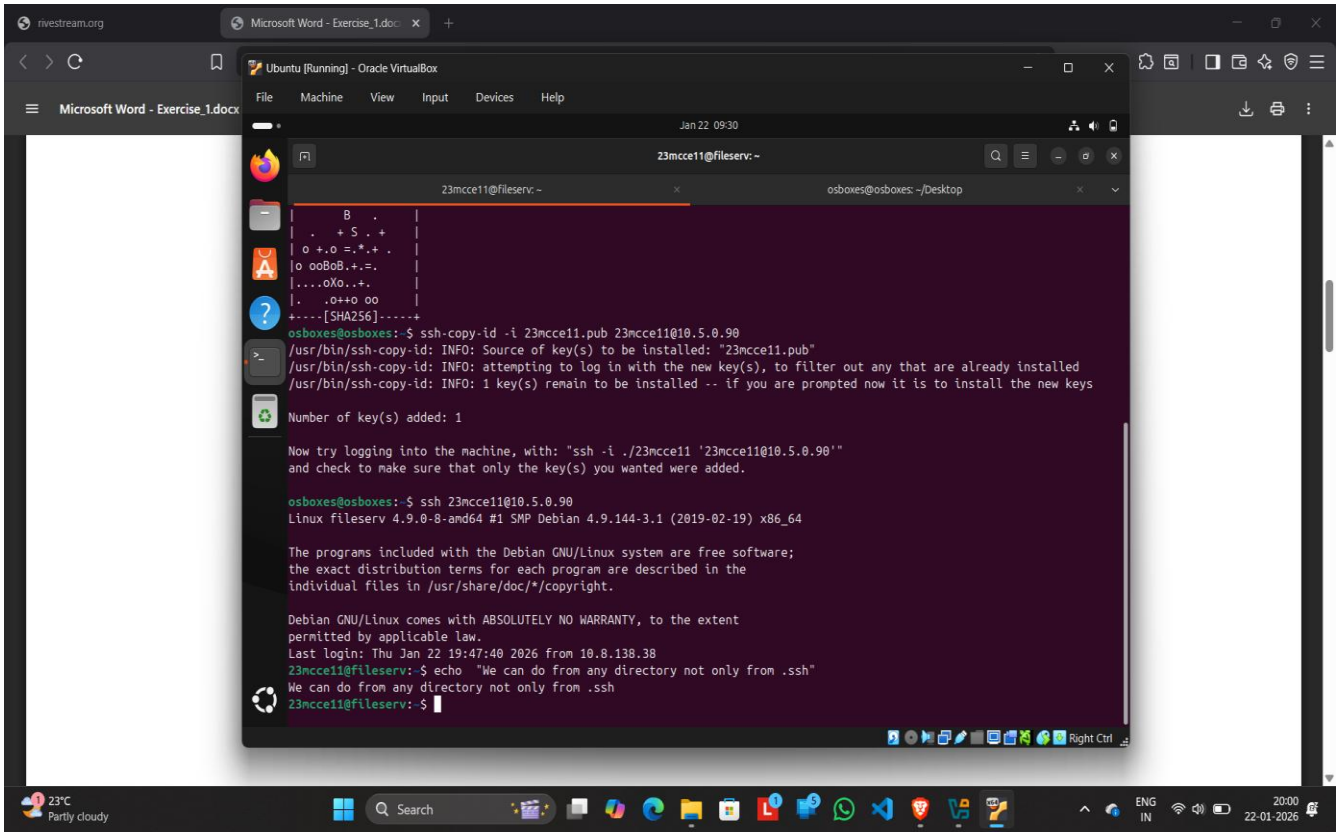
Q2 :

a)

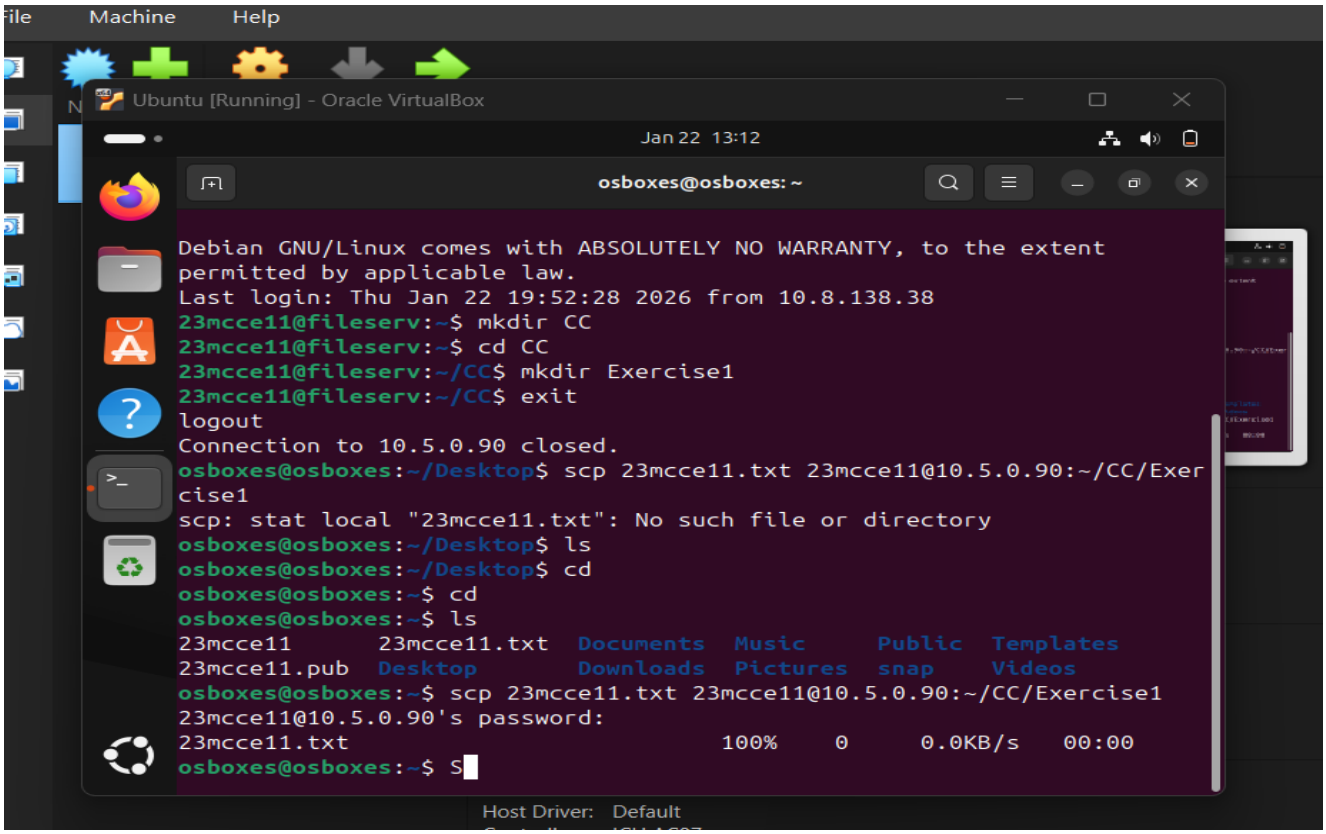

Ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jan 22 08:55
Type to search
23mcce11@fileserv: ~
osboxes@osboxes:~/Desktop$ 23mcce11@10.5.0.90
23mcce11@10.5.0.90: command not found
osboxes@osboxes:~/Desktop$ ssh 23mcce11@10.5.0.90
The authenticity of host '10.5.0.90 (10.5.0.90)' can't be established.
ED25519 key fingerprint is SHA256://9fXo7SOxSWqYDJAVGrF3zwLwrWlHvFxJAS8TmRywU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '10.5.0.90' (ED25519) to the list of known hosts.
23mcce11@10.5.0.90's password:
Linux fileserv 4.9.0-8-amd64 #1 SMP Debian 4.9.144-3.1 (2019-02-19) x86_64
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Thu Nov 13 00:36:41 2025 from 10.8.138.37
23mcce11@fileserv:~$
Mouse integration ...
Don't show again
Auto capture keyboard ...
Don't show again
Right Ctrl
Air: Moderate
Now
Search
ENG
IN
19:25
22-01-2026

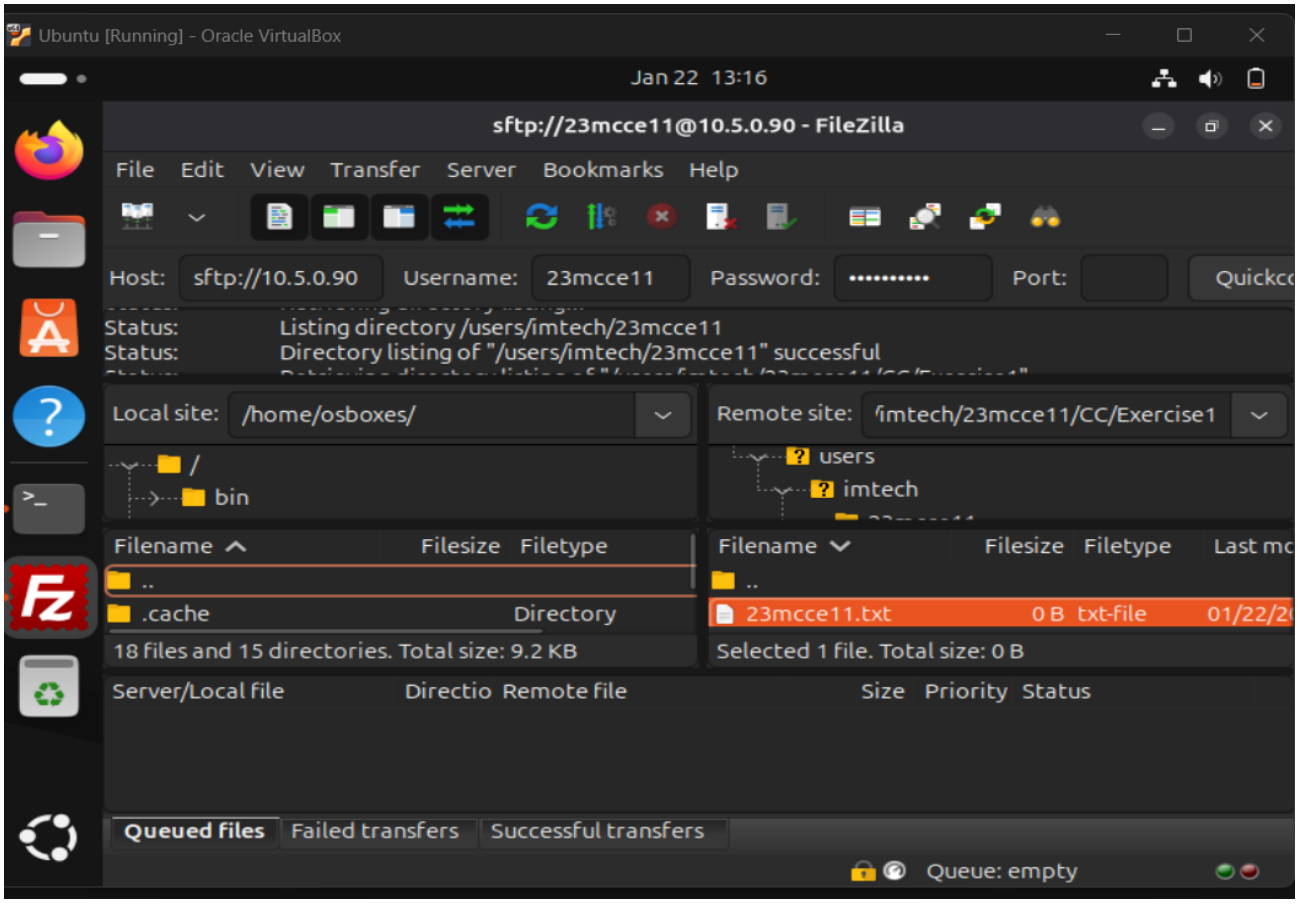
B)


Microsoft Word - Exercise_1.docx
Ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jan 22 09:28
23mcce11@fileserv: ~
23mcce11@fileserv: ~
osboxes@osboxes: ~/Desktop
osboxes@osboxes:~$ cd .ssh
osboxes@osboxes:~/.ssh$ ls
23mcce11 23mcce11.pub authorized_keys known_hosts known_hosts.old
osboxes@osboxes:~/.ssh$ rm 23mcce11 23mcce11.pub
osboxes@osboxes:~/.ssh$ ls
authorized_keys known_hosts known_hosts.old
osboxes@osboxes:~/.ssh$ cd ..
osboxes@osboxes:~$ ssh-keygen -t rsa
Generating public/private rsa key pair.
Enter file in which to save the key (/home/osboxes/.ssh/id_rsa): 23mcce11
Enter passphrase for "23mcce11" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in 23mcce11
Your public key has been saved in 23mcce11.pub
The key fingerprint is:
SHA256:8OpzdLuLlaqC9fgAi6nVM0dqN1rZTcejGLwQW5UHGMY osboxes@osboxes
The key's randomart image is:
+---[RSA 3072]----+
| .oooo |
| .E.. . |
| o . . |
| B . |
| . + S . + |
| o +.o =.*.+ . |
|o ooBoB.+.=. |
|....oXo..+. |
|. .o++o oo |
+----[SHA256]-----+
osboxes@osboxes:~$ ssh-copy-id -i 23mcce11.pub 23mcce11@10.5.0.90
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "23mcce11.pub"
Right Ctrl
23°C
Partly cloudy
Search
ENG
IN
19:58
22-01-2026

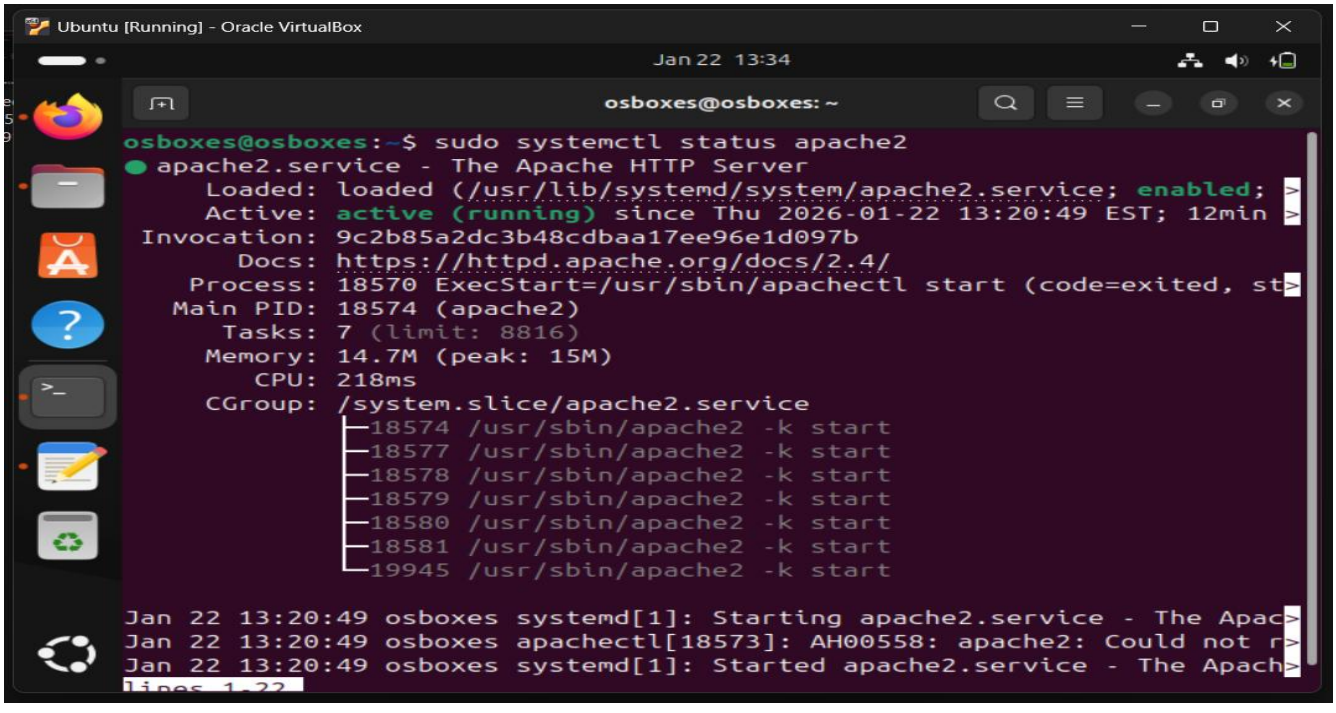

Ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jan 22 09:30
23mcce11@fileserv: ~
23mcce11@fileserv: ~
osboxes@osboxes: ~/Desktop
osboxes@osboxes:~$ ssh-copy-id -i 23mcce11.pub 23mcce11@10.5.0.90
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "23mcce11.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
Number of key(s) added: 1
Now try logging into the machine, with: "ssh -i ./23mcce11 '23mcce11@10.5.0.90'"
and check to make sure that only the key(s) you wanted were added.
osboxes@osboxes:~$ ssh 23mcce11@10.5.0.90
Linux fileserv 4.9.0-8-amd64 #1 SMP Debian 4.9.144-3.1 (2019-02-19) x86_64
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Thu Jan 22 19:47:40 2026 from 10.8.138.38
23mcce11@fileserv:~$ echo "We can do from any directory not only from .ssh"
We can do from any directory not only from .ssh
23mcce11@fileserv:~$
Right Ctrl
23°C Partly cloudy
Search
ENG IN
20:00 22-01-2026

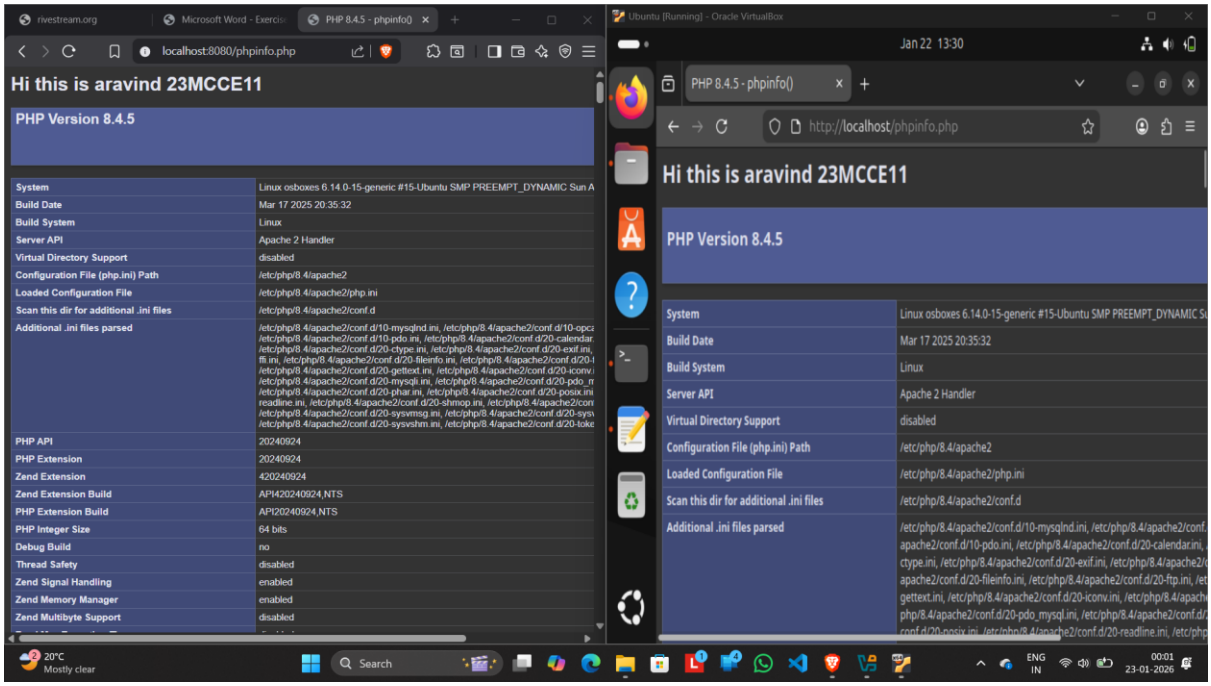
c)


Machine Help
Ubuntu [Running] - Oracle VirtualBox
Jan 22 13:12
osboxes@osboxes: ~
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Thu Jan 22 19:52:28 2026 from 10.8.138.38
23mcce11@fileserv:~$ mkdir CC
23mcce11@fileserv:~$ cd CC
23mcce11@fileserv:~/CC$ mkdir Exercise1
23mcce11@fileserv:~/CC$ exit
logout
Connection to 10.5.0.90 closed.
osboxes@osboxes:~/Desktop$ scp 23mcce11.txt 23mcce11@10.5.0.90:~/CC/Exer
cise1
scp: stat local "23mcce11.txt": No such file or directory
osboxes@osboxes:~/Desktop$ ls
osboxes@osboxes:~/Desktop$ cd
osboxes@osboxes:~$ cd
osboxes@osboxes:~$ ls
23mcce11 23mcce11.txt Documents Music Public Templates
23mcce11.pub Desktop Downloads Pictures snap Videos
osboxes@osboxes:~$ scp 23mcce11.txt 23mcce11@10.5.0.90:~/CC/Exercise1
23mcce11@10.5.0.90's password:
23mcce11.txt 100% 0 0.0KB/s 00:00
osboxes@osboxes:~$ S
Host Driver: Default


Ubuntu [Running] - Oracle VirtualBox
Jan 22 13:16
sftp://23mcce11@10.5.0.90 - FileZilla
File Edit View Transfer Server Bookmarks Help
Host: sftp://10.5.0.90 Username: 23mcce11 Password: Port:
Status: Listing directory /users/imtech/23mcce11
Status: Directory listing of "/users/imtech/23mcce11" successful
Local site: /home/osboxes/
Remote site: /imtech/23mcce11/CC/Exercise1
/
bin
users
imtech
Filename Filesize Filetype
..
.cache Directory
18 files and 15 directories. Total size: 9.2 KB
Filename Filesize Filetype Last mo
..
23mcce11.txt 0 B txt-file 01/22/2
Selected 1 file. Total size: 0 B
Server/Local file Directio Remote file Size Priority Status
Queued files Failed transfers Successful transfers
Queue: empty

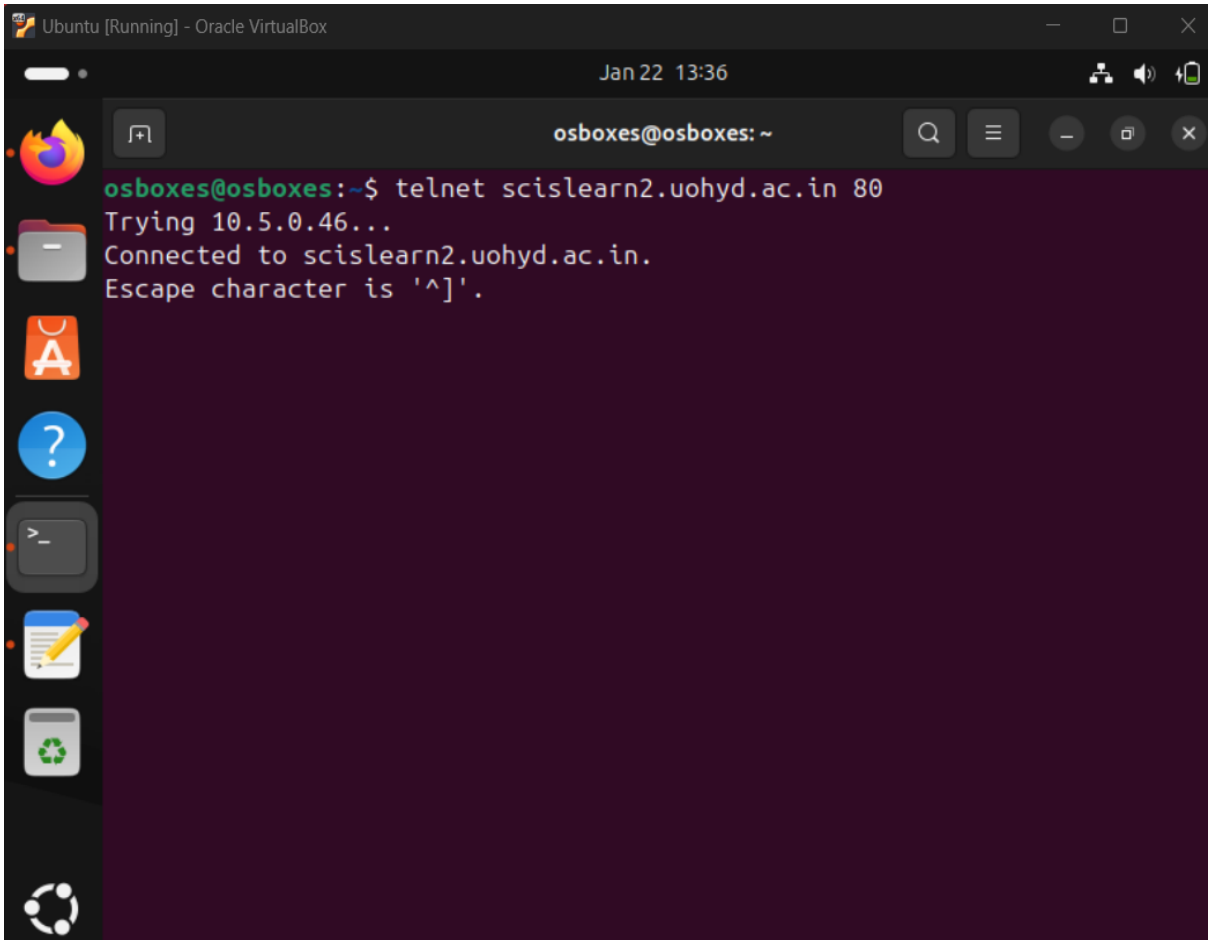
Q3 :


Ubuntu [Running] - Oracle VirtualBox
Jan 22 13:34
osboxes@osboxes: ~
osboxes@osboxes:~$ sudo systemctl status apache2
apache2.service - The Apache HTTP Server
Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled;
Active: active (running) since Thu 2026-01-22 13:20:49 EST; 12min
Invocation: 9c2b85a2dc3b48cdbaa17ee96e1d097b
Docs: https://httpd.apache.org/docs/2.4/
Process: 18570 ExecStart=/usr/sbin/apachectl start (code=exited, st
Main PID: 18574 (apache2)
Tasks: 7 (limit: 8816)
Memory: 14.7M (peak: 15M)
CPU: 218ms
CGroup: /system.slice/apache2.service
18574 /usr/sbin/apache2 -k start
18577 /usr/sbin/apache2 -k start
18578 /usr/sbin/apache2 -k start
18579 /usr/sbin/apache2 -k start
18580 /usr/sbin/apache2 -k start
18581 /usr/sbin/apache2 -k start
19945 /usr/sbin/apache2 -k start
Jan 22 13:20:49 osboxes systemd[1]: Starting apache2.service - The Apac
Jan 22 13:20:49 osboxes apachectl[18573]: AH00558: apache2: Could not r
Jan 22 13:20:49 osboxes systemd[1]: Started apache2.service - The Apach
lines 1-22

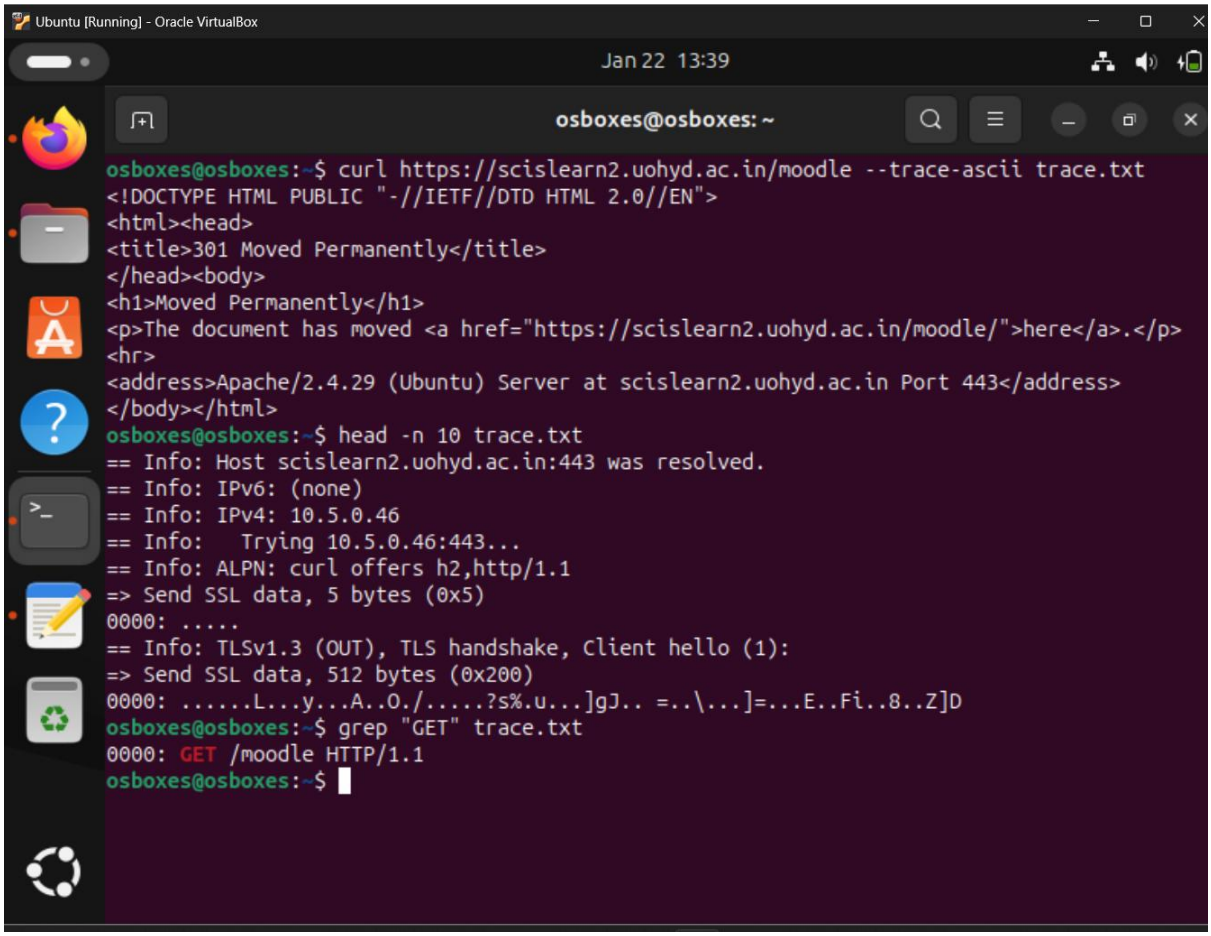

Hi this is aravind 23MCCE11
PHP Version 8.4.5
Hi this is aravind 23MCCE11
PHP Version 8.4.5

Q4 :

A)


Ubuntu [Running] - Oracle VirtualBox
Jan 22 13:36
osboxes@osboxes: ~
osboxes@osboxes:~$ telnet scislearn2.uohyd.ac.in 80
Trying 10.5.0.46...
Connected to scislearn2.uohyd.ac.in.
Escape character is '^]'.

b)



c) Request Headers (curl → Server)

curl sends these headers to tell the server what it wants:

Host - Which website (scislearn2.uohyd.ac.in)

User-Agent - "I'm curl program"

Accept - "Send any file type"

Accept-Encoding - "Compress files to save bandwidth"

Connection - "Keep connection open"

4d: Response Headers (Server → curl)

Server sends these headers with the webpage:

200 OK - "Request successful"

Content-Type - "This is HTML"

Content-Length - "File size: X bytes"

Server - "I'm Apache server"

Date - "Sent on [timestamp]"

Cache-Control - "Cache for 1 hour"

Request Headers (curl → Server): When curl makes a request, it sends headers like Host (which website: scislearn2.uohyd.ac.in), User-Agent ("I'm curl program"), Accept ("send any file type"), Accept-Encoding ("compress files to save bandwidth"), Connection ("keep connection open"), Authorization (API login tokens), Content-Type ("I'm sending JSON"), Content-Length (data size in bytes), and Cookie (stay logged in). These tell the server exactly what curl wants and how to respond.

d) The response headers are:

1)status line:

->this isnt technically a header,but its the most important line.it confirms the protocol and the success code

2)date:

->the exact date and time the server sent the response.this is a crucial for caching

3)server:

->it tells us what software it is running examples are apache and nginx

4)content-type:

->this tells browser how to render the data

5)content-length:

->the size of the in bytes

6)connection:

->tells the computer to keep te tcp connection open after the request is finished,so subsequent request are faster

7)last-modified:

->tells us when the content of the page was last modified